TEXORA · COURSE NOTES · LECTURE SHEET

Differential Calculus of Several Variables

Functions of two or more variables — partial derivatives, the total differential, the chain rule, Euler's theorem, and maxima and minima.

MATH-1101 · MATHEMATICS-I · TOPIC 4 · LEVEL 1 TERM 1

1. Functions of several variables

Most real quantities depend on several inputs at once: the volume of a gas on pressure and temperature, the strength of a yarn on fibre length, fineness and twist. A function $z = f(x, y)$ assigns a height to each point (x, y) of its domain, and its graph is a surface. Limits and continuity extend from the one-variable theory, with one honest difficulty: a point in the plane can be approached along infinitely many paths, and the limit exists only when every path gives the same value.

2. Partial derivatives

The **partial derivative** of f with respect to x measures the rate of change of f as x varies *while y is held constant*:

$$\partial f/\partial x = \lim_{h \to 0} [\, f(x + h, y) - f(x, y) \,] / h,$$

and $\partial f/\partial y$ is defined symmetrically. In practice, differentiate with respect to the chosen variable treating all others as constants. Standard notation: f_x, f_y for first order; f_{xx}, f_{yy}, f_{xy}, f_{yx} for second order — and for well-behaved functions the mixed derivatives are equal, $f_{xy} = f_{yx}$ (Clairaut's theorem). Geometrically, f_x and f_y are the slopes of the surface in the x and y directions; together they determine the tangent plane at a point.

2.1 The total differential

When both variables change a little, the function changes by approximately

$$dz = f_x\, dx + f_y\, dy,$$

the **total differential**. It is the several-variable version of the tangent-line approximation, and the standard tool for estimating small errors: if x and y are measured with small errors dx and dy, then dz

estimates the resulting error in z.

2.2 The chain rule

If $z = f(x, y)$ and x and y are themselves functions of t, then z changes with t through both channels:

$$dz/dt = f_x \, dx/dt + f_y \, dy/dt.$$

WORKED EXAMPLE — CHAIN RULE, CHECKED DIRECTLY

Let $z = x^2y$ with $x = t^2$, $y = t^3$. By the chain rule: $dz/dt = (2xy)(2t) + (x^2)(3t^2) = (2 \cdot t^2 \cdot t^3)(2t) + (t^4)(3t^2) = 4t^6 + 3t^6 = 7t^6$. Directly, $z = t^4 \cdot t^3 = t^7$, so $dz/dt = 7t^6$ ✓ — the two routes agree.

3. Euler’s theorem on homogeneous functions

A function f(x, y) is **homogeneous of degree n** when scaling both variables scales the value by the n-th power: $f(tx, ty) = t^n f(x, y)$. Euler’s theorem connects this to the partial derivatives:

$$x \, f_x + y \, f_y = n \, f.$$

WORKED EXAMPLE — VERIFYING EULER’S THEOREM

$f(x, y) = x^2 + xy$ is homogeneous of degree 2 (every term has total degree 2). $f_x = 2x + y$ and $f_y = x$, so $x \, f_x + y \, f_y = x(2x + y) + y(x) = 2x^2 + 2xy = 2f$ ✓, as the theorem requires with n = 2.

4. Maxima and minima of two variables

At an interior maximum or minimum (a **stationary point**), both partial derivatives vanish: $f_x = 0$ and $f_y = 0$. The **second derivative test** then classifies the point using the discriminant $D = f_{xx} \, f_{yy} - f_{xy}^2$:

TEST RESULT	NATURE OF THE STATIONARY POINT
$D > 0$ and $f_{xx} > 0$	Local minimum
$D > 0$ and $f_{xx} < 0$	Local maximum
$D < 0$	Saddle point (neither max nor min)
$D = 0$	Test inconclusive

WORKED EXAMPLE — CLASSIFYING A STATIONARY POINT

$f(x, y) = x^2 + y^2 - 4x + 6y$. $f_x = 2x - 4 = 0$ gives $x = 2$; $f_y = 2y + 6 = 0$ gives $y = -3$. Second derivatives: $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$, so $D = 4 > 0$ with $f_{xx} > 0$ — a **local minimum** at $(2, -3)$, with

value $f = 4 + 9 - 8 - 18 = \mathbf{-13}$.

4.1 Constrained problems: Lagrange multipliers

When the variables are tied by a constraint $g(x, y) = c$ (a fixed budget, a fixed perimeter, a fixed material quantity), extrema of f on the constraint occur where $\nabla f = \lambda \nabla g$ for some multiplier λ — that is, where the level curves of f and g touch. Solve the system $f_x = \lambda g_x$, $f_y = \lambda g_y$, $g(x, y) = c$ together. This is the mathematical backbone of optimisation under a fixed resource.

5. Two further tools

- **Jacobian.** For a change of variables $(x, y) \to (u, v)$, the Jacobian determinant $J = \partial(x, y)/\partial(u, v) = x_u\, y_v - x_v\, y_u$ measures local area scaling; it reappears decisively in multiple integrals (Topic 5), where it converts dA between coordinate systems.
- **Taylor's series for two variables** extends the one-variable expansion: $f(x + h, y + k) \approx f + (h\, f_x + k\, f_y) + \tfrac{1}{2}(h^2 f_{xx} + 2hk\, f_{xy} + k^2 f_{yy}) + \cdots$, evaluated at (x, y). Its first-order part is exactly the total differential; the second-order part drives the maxima–minima test above.

6. The tangent plane — the total differential at work

The total differential has a picture: at a point (a, b) on the surface $z = f(x, y)$, the plane $z = f(a, b) + f_x(a, b)\,(x - a) + f_y(a, b)\,(y - b)$ touches the surface and matches both its slopes — the **tangent plane**. For $f(x, y) = x^2 + y^2$ at $(1, 2)$: $f = 5$, $f_x = 2$, $f_y = 4$, so the tangent plane is $z = 5 + 2(x - 1) + 4(y - 2)$. Reading it as an approximation: near $(1, 2)$, $f(1.1, 1.9) \approx 5 + 2(0.1) + 4(-0.1) = 4.8$, against the exact $1.21 + 3.61 = 4.82$ — the differential doing its everyday job of estimating change without recomputing the function.

7. Key points and summary

- Partial derivatives differentiate with respect to one variable while the rest are held constant; mixed partials are equal for well-behaved functions.
- The total differential $dz = f_x\, dx + f_y\, dy$ accumulates both channels of change, and the chain rule divides change along any path through those channels.
- Euler's theorem, $x f_x + y f_y = n f$, holds exactly for functions homogeneous of degree n.
- Stationary points satisfy $f_x = f_y = 0$ and are classified by $D = f_{xx} f_{yy} - f_{xy}^2$; constraints are handled by Lagrange multipliers, and the Jacobian prepares for coordinate changes in integration.

8. Practice

Work each problem fully; final answers are given at the end.

1. For $f(x, y) = x^2y + 3y^2$, find f_x and f_y.
2. Estimate dz for $z = x^2 + y^2$ at (1, 2) when $dx = 0.1$ and $dy = 0.2$.
3. Verify Euler's theorem for $f(x, y) = x^3 + y^3 + xy^2$, stating the degree first.
4. Classify the stationary point of $f(x, y) = x^2 + y^2 + 4x - 2y$ and give the extreme value.
5. For $z = e^{xy}$ with $x = t$, $y = t^2$, find dz/dt by the chain rule and check by substitution.

Answers

1. $f_x = 2xy$; $f_y = x^2 + 6y$.
2. $dz = 2x\,dx + 2y\,dy = 0.2 + 0.8 = 1.0$.
3. Degree 3. $f_x = 3x^2 + y^2$, $f_y = 3y^2 + 2xy$; $x f_x + y f_y = 3x^3 + 3y^3 + 3xy^2 = 3f$ ✓.
4. Stationary point $(-2, 1)$; $D = 4 > 0$ with $f_{xx} = 2 > 0 \rightarrow$ local minimum, value -5.
5. $dz/dt = y\,e^{xy} \cdot 1 + x\,e^{xy} \cdot 2t = 3t^2\,e^{t^3}$; by substitution $z = e^{t^3}$, whose derivative is the same ✓.
